

Naveen Sita Reddy Kovvuri

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SUMMARY

Adaptable **Software Engineer** with expertise in **Python development**, **API testing**, and **automation frameworks** (Selenium, Pytest, Postman). Skilled in **web development** (HTML5, CSS3, Flask), **cloud computing** with **AWS**, **microservices architecture** and **CI/CD pipelines** (Git, Jenkins). Proficient in **SQL**, **PL/SQL**, and **database testing**, with experience in **data science** and **machine learning** (Pandas, NumPy, Scikit-learn). Strong foundation in **data structures and algorithms**, enabling efficient problem-solving and optimized solutions. Proven track record of delivering solutions at **TCS**, thriving in **agile environments** and contributing to collaborative team projects.

TECHNICAL SKILLS

- **Programming Languages and Scripting:** Python (including Python 3, Python Scripting and Automation, Game Development, GUI with Tkinter and Turtle), Java, Swift, Bash Command Line, JavaScript/TypeScript, Go(Familiar)
- **Web Development:** Front-End: HTML5, CSS3, Bootstrap 4, UX/UI Design, Back-End: Flask, RESTful APIs, WTForms, Authentication, Full-Stack Deployment: GitHub Pages, Heroku, GUnicorn
- **DevOps and Version Control:** Git, GitHub, Jenkins, Docker(Basics)
- **Cloud Platforms:** AWS (EC2, S3, RDS, SageMaker), GCP(Basics)
- **Testing and QA:** Manual Testing, Regression Testing, Automation Testing (Selenium WebDriver, Pytest, BDD with Behave), API Testing (Postman), Database Testing, Unit Testing, Integration Testing, Testing Methodologies, Software Debugging Tools, Cross-Functional Testing, Test Plans, Software Quality Assurance, JIRA, TestRail
- **Database Management:** PL/SQL, SQL, SQLite, PostgreSQL
- **Data Science and Analysis:** Pandas, NumPy, Matplotlib, Plotly, Seaborn, Scikit-learn
- **Tools and IDEs:** PyCharm, Visual Studio Code, Jupyter Notebook, Google Colab
- **Operating Systems:** MacOS, IOS, Windows, KaliLinux

EXPERIENCE

- **Assistant System Engineer** Dec 2021 – Dec 2022
Tata Consultancy Services, West Bengal, India
Tools: Jenkins, Git, Postman, PL/SQL, JIRA, SQL
 - Collaborated with stakeholders in the **Banking and Financial Services (BFS)** domain to gather and analyze **requirements**, ensuring alignment with project goals.
 - Conducted **manual testing** for banking applications, including **functional**, **regression**, and **integration testing** to ensure seamless user experiences.
 - Managed **CI/CD pipelines** for banking applications using **Jenkins** and **Git** for **version control**, improving delivery time by 30%.
 - Performed **API testing** using **Postman** to validate the functionality, reliability, and performance of banking application interfaces.
 - Collaborated with **Java**-based backend teams to integrate services and troubleshoot interface issues.
 - Executed **database testing** with **PL/SQL**, validating data **accuracy**, **integrity**, and **security** of financial data across various systems.
 - Provided **production support**, troubleshooting issues, investigating root causes, and ensuring minimal downtime for banking applications.
 - Utilized **JIRA** for **bug tracking** and **task management**, ensuring transparent communication between **cross-functional teams**.
 - Collaborated with **developers** and **project managers** to ensure timely delivery and high **quality** of project milestones.

PROJECTS

- **Emotion Based Music Player** [GitHub] Mar 2025 – Apr 2025
Tools: TensorFlow, OpenCV, Machine Learning, Django, CNN, HTML/CSS, JavaScript
 - Developed an **AI-integrated** web application that uses **facial emotion recognition** to personalize music playlists and animated UI themes.
 - Implemented a **Convolutional Neural Network (CNN)** using **TensorFlow** to detect emotional states (Angry, Happy, Calm, Sad) from real-time webcam input.
- **Python Interpreter** [GitHub] Oct 2024 – Dec 2024
Tools: Python, Lex, Yacc, OOP
 - Developed a **Python interpreter** for parsing and executing basic Python programs with **Lex** and **Yacc**.
 - Implemented **error handling**, **expression evaluation**, and support for **loops**, **arithmetic operations**.
- **Anomaly Detection for Financial Data** [GitHub] Nov 2024 – Dec 2024
Tools: Python, Pandas, Numpy, Matplotlib, Seaborn, Scikit-learn
 - Developed an anomaly detection system using Python, Scikit-learn, and statistical modeling to identify irregularities in financial data with high precision, leveraging Isolation Forest, One-Class SVM, and EDA techniques.

EDUCATION

Southeast Missouri State University <i>Master of Science in Applied Computer Science GPA: 3.7/4.0</i>	Cape Girardeau, Missouri, USA <i>Aug 2023 – May 2025</i>
Jawaharlal Nehru Technological University <i>Bachelor of Technology in Electronics and Communication Engineering Percentage: 71%</i>	Kakinada, Andhra Pradesh, India <i>July 2016 – May 2020</i>

CERTIFICATIONS

- **AWS Certified Solutions Architect – Associate (In Progress)** *Amazon Web Services (AWS)*
Preparing for associate-level certification focusing on designing scalable, resilient, and secure applications on the AWS cloud platform. Covers core AWS services, architecture best practices, and cost-optimized solutions. Also exploring **GCP** fundamentals.
Skills: VPC, EC2, S3, IAM, RDS, Load Balancing, Auto Scaling, Cloud Security, High Availability, Cost Optimization.
- **Certification in Swift Programming** *BRN InfoTech, India*
Hands-on training in Apple's Swift programming language with an introduction to iOS development. Covered control flow, optionals, classes, and functions.
Skills: Swift, Xcode, iOS Fundamentals, iOS, macOS, Control Structures, Object-Oriented Programming
- **UX/UI Design with Adobe XD** *Udemy*
Completed project-based learning in modern UX/UI design using Adobe XD. Designed interactive wireframes, created prototypes, and mapped user journeys.
Skills: Adobe XD, Wireframing, Prototyping, UI Components, User Flow Design, Visual Hierarchy
- **Web Development – HTML and CSS** *Udemy*
Built mobile-responsive web pages using semantic HTML5 and CSS3 with Flexbox and media queries. Applied design principles for layout and accessibility.
Skills: HTML5, CSS3, Responsive Design, Flexbox, Media Queries, Page Structure, Basic SEO
- **C Programming Certification** *IIT Bombay – Spoken Tutorial*
Learned the fundamentals of C language with an emphasis on syntax, logic building, and memory management. Created modular programs using functions and pointers.
Skills: C Programming, Pointers, Arrays, Functions, Memory Allocation, Loops
- **Python Programming** *Spoken Tutorial (Govt. of India)*
Developed a solid understanding of Python programming with advanced topics such as object-oriented programming (OOP), modules, and libraries.
Skills: Python, Object-Oriented Programming (OOP), Libraries (NumPy, Pandas), Data Structures, File I/O, Error Handling